

Diagnostic information - DME (DFI) control unit

Catalytic converter, bank 1 - inadequate effect

INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

Diagnostic conditions

- Perform all short tests in sequence

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °F)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

- If an oxygen sensor fault is entered:
 - ♦ Correct the cause of the oxygen sensor fault according to instructions, then perform short test as described above and check whether the catalytic converter fault is entered again
- ♦ Exhaust system leak between engine and rear silencer
- ♦ Oxygen sensor upstream and downstream of catalytic converter bank 1 interchanged
- ♦ Oxygen sensor upstream of catalytic converter bank 1 and bank 2 interchanged
- ♦ Oxygen sensor downstream of catalytic converter bank 1 and bank 2 interchanged
- ♦ Catalytic converter bank 1 faulty

Diagnostic information - DME (DFI) control unit

Catalytic converter, bank 2 - inadequate effect

INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

Diagnostic conditions

- Perform all short tests in sequence

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °F)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

- If an oxygen sensor fault is entered:
 - ♦ Correct the cause of the oxygen sensor fault according to instructions, then perform short test as described above
- ♦ Exhaust system leak between engine and rear silencer
- ♦ Oxygen sensor upstream and downstream of catalytic converter bank 2 interchanged
- ♦ Oxygen sensor upstream of catalytic converter bank 1 and bank 2 interchanged
- ♦ Oxygen sensor downstream of catalytic converter bank 1 and bank 2 interchanged
- ♦ Catalytic converter bank 2 faulty

P043E

Diagnostic information - DME (DFI)

DMTL - Tank leakage diagnosis module

Diagnostic conditions

- Perform tank leak short test

or

1. Meet preconditions:

- No faults for ambient pressure sensor, coolant temperature sensor, vehicle speed sensor
- No faults on DMTL module or tank vent valve
- Ignition off before engine start for at least 6 hours
- Engine coolant temperature (T020) above 4 °C (39 °F)

2. Then engine running for at least 10 minutes without interruption under the following conditions:

- Ambient temperature (T052) over 4 ... 38 °C (39 ... 100 °F)
- Ambient pressure (A060) over 730 hPa (location below 2,700 m (8,858 ft) above sea level)
- Engine coolant temperature at engine start (T072) above 4 °C (39 °F)

3. Then ignition switched off for more than 12 minutes under the following conditions:

- Consistent vehicle voltage of 11 to 16 V
- NO battery charger connected
- Nothing actuated on the vehicle (opening doors, switching on loads, etc.)
- No refueling process

Possible fault causes

- ♦ DMTL - tank leakage diagnosis module faulty

P043F

Diagnostic information - DME (DFI)

DMTL - Tank leakage diagnosis module

Diagnostic conditions

- Perform tank leak short test

or

1. Meet preconditions:

- No faults for ambient pressure sensor, coolant temperature sensor, vehicle speed sensor
- No faults on DMTL module or tank vent valve
- Ignition off before engine start for at least 6 hours
- Engine coolant temperature (T020) above 4 °C (39 °F)

2. Then engine running for at least 10 minutes without interruption under the following conditions:

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- Consistent vehicle voltage of 11 to 16 V
- NO battery charger connected
- Nothing actuated on the vehicle (opening doors, switching on loads, etc.)
- No refueling process

Possible fault causes

- ♦ DMTL - tank leakage diagnosis module faulty

Diagnostic information - DME (DFI) control unit

Tank vent system lower partial load

Diagnostic conditions

- Perform Tank vent valve short test - lower partial-load range

OR

- Time since engine start > 2 minute(s)
- Idling > 15 second(s)
- Engine temperature > 40 °C (104 °F)
- Ambient temperature > 4 °C (39 °F)
- Ambient pressure > 730 hPa (= location below approx 2,400 ... 2,700 m/7,874 ... 8,858 ft) (A060)
- All oxygen sensor(s) ready for operation
- None of the following faults stored: Oxygen sensor(s), tank vent valve output stage, intake manifold pressure sensor, tank ventilation pressure sensor
- Vehicle speed > 15 km/h (9 mph)
- The carbon canister is adequately flushed via the tank vent valve (engine idling).

Possible fault causes

- ♦ Tank vent valve faulty (stuck open or closed)
- ♦ Tank vent valve not installed or installed incorrectly
- ♦ Line between active carbon filter and tank vent valve leaking/trapped/blocked
- ♦ Purge-air line between tank vent valve and intake manifold leaking/trapped/blocked
- ♦ Check valve between tank vent valve and intake manifold faulty or installed backward
- ♦ Check valve between tank vent valve and turbocharger (Venturi pump) faulty or installed backward

Fault setting condition(s)

- The fault is set if the pressure difference between the ambient pressure and the pressure in the tank vent line is too low when the tank vent valve is activated (depending on the respective engine operating point).

Diagnostic information - DME (DFI)

Tank system micro leak

Diagnostic conditions

- Perform tank leak short test

or

1. Meet preconditions:

- No faults for ambient pressure sensor, coolant temperature sensor, vehicle speed sensor
- No faults on DMTL module or tank vent valve
- Ignition off before engine start for at least 6 hours
- Engine coolant temperature (T020) above 4 °C (39 °F)

2. Then engine running for at least 10 minutes without interruption under the following conditions:

- Ambient temperature (T052) over 4 ... 38 °C (39 ... 100 °F)
- Ambient pressure (A060) over 730 hPa (location below 2,700 m (8,858 ft) above sea level)
- Engine coolant temperature at engine start (T072) above 4 °C (39 °F)

3. Then ignition switched off for more than 12 minutes under the following conditions:

- Consistent vehicle voltage of 11 to 16 V
- NO battery charger connected
- Nothing actuated on the vehicle (opening doors, switching on loads, etc.)
- No refueling process

Possible fault causes

- ♦ Tank cap not closed correctly, leaking or missing
- ♦ Purge-air line leaking
- ♦ Tank vent valve leaking
- ♦ Tank leakage diagnosis module (DMTL) leaking
- ♦ Leak in fuel tank system

P0444

Diagnostic information - DME (DFI)

Tank vent valve driver

Diagnostic conditions

- Vehicle voltage > 9 V
- Engine start detected

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Tank vent valve faulty

Diagnostic information - DME (DFI)

DMTL - Tank Leakage Diagnostics Module, valve driver

Diagnostic conditions

- Run [DMTL valve](#) drive link

or

- Perform tank leak short test

or

1. Meet preconditions:

- No faults for ambient pressure sensor, coolant temperature sensor, vehicle speed sensor
- No faults on DMTL module or tank vent valve
- Ignition off before engine start for at least 6 hours
- Engine coolant temperature ([T020](#)) above 4 °C (39 °F)

2. Then engine running for at least 10 minutes without interruption under the following conditions:

- Ambient temperature ([T052](#)) over 4 ... 38 °C (39 ... 100 °F)
- Ambient pressure ([A060](#)) over 730 hPa (location below 2,700 m (8,858 ft) above sea level)
- Engine coolant temperature at engine start ([T072](#)) above 4 °C (39 °F)

3. Then ignition switched off for more than 12 minutes under the following conditions:

- Consistent vehicle voltage of 11 to 16 V
- NO battery charger connected
- Nothing actuated on the vehicle (opening doors, switching on loads, etc.)
- No refueling process

Possible fault causes

- ♦ Open circuit, short circuit to B+ or short circuit to ground in activation line to change-over valve for DMTL module
- ♦ Open circuit in the power supply to the DMTL module
- ♦ DMTL - tank leakage diagnosis module faulty

Diagnostic information - DME (DFI)

DMTL - Tank Leakage Diagnostics Module, valve driver

Diagnostic conditions

- Run [DMTL valve](#) drive link

or

- Perform tank leak short test

or

1. Meet preconditions:

- No faults for ambient pressure sensor, coolant temperature sensor, vehicle speed sensor
- No faults on DMTL module or tank vent valve
- Ignition off before engine start for at least 6 hours
- Engine coolant temperature ([T020](#)) above 4 °C (39 °F)

2. Then engine running for at least 10 minutes without interruption under the following conditions:

- Ambient temperature ([T052](#)) over 4 ... 38 °C (39 ... 100 °F)
- Ambient pressure ([A060](#)) over 730 hPa (location below 2,700 m (8,858 ft) above sea level)
- Engine coolant temperature at engine start ([T072](#)) above 4 °C (39 °F)

3. Then ignition switched off for more than 12 minutes under the following conditions:

- Consistent vehicle voltage of 11 to 16 V
- NO battery charger connected
- Nothing actuated on the vehicle (opening doors, switching on loads, etc.)
- No refueling process

Possible fault causes

- ♦ Open circuit, short circuit to B+ or short circuit to ground in activation line to change-over valve for DMTL module
- ♦ Open circuit in the power supply to the DMTL module
- ♦ DMTL - tank leakage diagnosis module faulty

Diagnostic information - DME (DFI)

Tank system micro leak

Diagnostic conditions

- Perform tank leak short test

or

1. Meet preconditions:

- No faults for ambient pressure sensor, coolant temperature sensor, vehicle speed sensor
- No faults on DMTL module or tank vent valve
- Ignition off before engine start for at least 6 hours
- Engine coolant temperature (T020) above 4 °C (39 °F)

2. Then engine running for at least 10 minutes without interruption under the following conditions:

- Ambient temperature (T052) over 4 ... 38 °C (39 ... 100 °F)
- Ambient pressure (A060) over 730 hPa (location below 2,700 m (8,858 ft) above sea level)
- Engine coolant temperature at engine start (T072) above 4 °C (39 °F)

3. Then ignition switched off for more than 12 minutes under the following conditions:

- Consistent vehicle voltage of 11 to 16 V
- NO battery charger connected
- Nothing actuated on the vehicle (opening doors, switching on loads, etc.)
- No refueling process

Possible fault causes

- ♦ Tank cap not closed correctly, leaking or missing
- ♦ Purge-air line leaking
- ♦ Tank vent valve leaking
- ♦ Tank leakage diagnosis module (DMTL) leaking
- ♦ Leak in fuel tank system

P0458

Diagnostic information - DME (DFI)

Tank vent valve driver

Diagnostic conditions

- Vehicle voltage > 9 V
- Engine start detected

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Tank vent valve faulty

P0459

Diagnostic information - DME (DFI)

Tank vent valve driver

Diagnostic conditions

- Vehicle voltage > 9 V
- Engine start detected

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Tank vent valve faulty

P0464

Diagnostic information - DME (DFI) control unit

Fuel level sensor

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ See Instrument cluster fault memory

Diagnostic information - DME (DFI) control unit

Radiator fan control unit driver

⚠ WARNING

Exposed running fan blades!

Danger of limbs being severed!

Risk of serious injury due to ejected foreign bodies!

- ♦ **ALWAYS** keep hair or clothing out of the rotating range of the fan.
- ♦ DO NOT reach into the rotating fan.
- ♦ **DO NOT** place tools or other items in the rotating range of the fan.
- ♦ Perform work on the fan only with the ignition switched off.
- ♦ Secure the ignition to prevent it from being switched on accidentally by other persons.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit to ground or open circuit in lines (power supply to radiator fan control unit terminal 30)
- ♦ Short circuit to B+/ground or open circuit in the line (radiator fan control unit activation)
- ♦ Radiator fan control unit faulty

Diagnostic information - DME (DFI) control unit

Driver for radiator fan control unit 2

⚠ WARNING

Exposed running fan blades!

Danger of limbs being severed!

Risk of serious injury due to ejected foreign bodies!

- ♦ **ALWAYS** keep hair or clothing out of the rotating range of the fan.
- ♦ DO NOT reach into the rotating fan.
- ♦ **DO NOT** place tools or other items in the rotating range of the fan.
- ♦ Perform work on the fan only with the ignition switched off.
- ♦ Secure the ignition to prevent it from being switched on accidentally by other persons.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit to ground or open circuit in lines (power supply to radiator fan control unit terminal 30)
- ♦ Short circuit to B+/ground or open circuit in the line (radiator fan control unit activation)
- ♦ Radiator fan control unit faulty

Diagnostic information - DME (DFI) control unit

Bank 1 secondary-air system

Diagnostic conditions

- Perform all short tests from top to bottom, up to and including secondary air.
- or
- The diagnostic function starts under the following conditions:
 - Engine running, with:
 - Ambient temperature > 5 °C (41 °F)
 - Engine start temperature 5 ... 32 °C (41 ... 89.6 °F)
 - Secondary-air injection active
 - None of the following faults stored: Oxygen sensor upstream of catalytic converter, engine coolant temperature sensor, camshaft adjustment, ambient pressure, ambient temperature
- The diagnostic function ends under the following conditions:
 - Oxygen sensor upstream of catalytic converter not ready for operation
 - Total time period (diagnostics) exceeded (depending on: engine start temperature)
- Diagnostics is interrupted for as long as the following conditions exist:
 - Engine speed > 3,500 rpm
 - Mass air flow > 750 mg/stroke
 - Overrun phase(s)

INFORMATION

- ♦ The diagnostic function ends with the end of secondary-air injection.
- ♦ A certain number of measurements must have taken place within diagnostics, between operational readiness of the oxygen sensors and the end of secondary-air injection.
- ♦ If this is not the case, diagnostics is ended without a result.

Possible fault causes

- ♦ Catalytic converter faulty (broken)
- ♦ Secondary-air injection line leaking or constricted
- ♦ Bank 1 secondary-air valve stuck/frozen
- ♦ Secondary-air duct on engine constricted/coked
- ♦ Fuel-air mixture preparation not adapted

Fault setting condition(s)

- Lambda control actual value <> setpoint deviation
 - The diagnostic function checks whether deviation of the lambda controllers is as expected upon activation.

Diagnostic information - DME (DFI) control unit

Bank 2 secondary-air system

Diagnostic conditions

- Perform all short tests from top to bottom, up to and including secondary air.
- or
- The diagnostic function starts under the following conditions:
 - Engine running, with:
 - Ambient temperature > 5 °C (41 °F)
 - Engine start temperature 5 ... 32 °C (41 ... 89.6 °F)
 - Secondary-air injection active
 - None of the following faults stored: Oxygen sensor upstream of catalytic converter, engine coolant temperature sensor, camshaft adjustment, ambient pressure, ambient temperature
- The diagnostic function ends under the following conditions:
 - Oxygen sensor upstream of catalytic converter not ready for operation
 - Total time period (diagnostics) exceeded (depending on: engine start temperature)
- Diagnostics is interrupted for as long as the following conditions exist:
 - Engine speed > 3,500 rpm
 - Mass air flow > 750 mg/stroke
 - Overrun phase(s)

INFORMATION

- ♦ The diagnostic function ends with the end of secondary-air injection.
- ♦ A certain number of measurements must have taken place within diagnostics, between operational readiness of the oxygen sensors and the end of secondary-air injection.
- ♦ If this is not the case, diagnostics is ended without a result.

Possible fault causes

- ♦ Catalytic converter faulty (broken)
- ♦ Secondary-air injection line leaking or constricted
- ♦ Bank 1 secondary-air valve stuck/frozen
- ♦ Secondary-air duct on engine constricted/coked
- ♦ Fuel-air mixture preparation not adapted

Fault setting condition(s)

- Lambda control actual value <> setpoint deviation
 - The diagnostic function checks whether deviation of the lambda controllers is as expected upon activation.

Diagnostic information - DME (DFI) control unit

Mass air flow in near idle speed range, implausibly high (fast deviation)

Diagnostic conditions

- Idling > 30 s.
 - Then engine speed > 1,500 rpm > 1 s.
 - Then idling > 30 s.
- None of the following faults stored: Intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Possible fault causes

- ♦ Oil filler cap open
- ♦ Major leak in the vacuum system (e.g. brake booster) because the vacuum pump evacuates the system towards the cylinder head
- ♦ Positive crankcase ventilation hose leaking
- ♦ Check valve on full-load positive crankcase ventilation hose (ahead of turbocharger) leaking
- ♦ Leak in intake system - between throttle valve and engine (intake valves)
- ♦ Tank vent valve faulty (stuck open)
- ♦ Intake manifold pressure sensor - electric plug connection loose, oxidized, corroded, interrupted
- ♦ Fault in areas: intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- Excessive deviation of the throttle valve position during the current engine idling phase compared to the most recent and the penultimate engine idling phases.

The diagnostic function checks whether the value

[H040-Mass air flow correction factor](#) deviates too much from the value during the last engine idling phase:

- ♦ If NO, the new value is used as the next reference value, and diagnostics begins again in the next engine idling phase.
- ♦ If YES, the test is repeated in the next engine idling phase and compared with the value of the penultimate engine idling phase:
 - ♦ If NO relevant deviation still exists, the new value is used as the next reference value.
 - ♦ If a relevant deviation is present AGAIN, the fault is entered.

P04F0

Diagnostic information - DME (DFI) control unit

Tank vent system upper partial load

Diagnostic conditions

- Perform Tank vent valve short test - upper partial-load range

OR

- Time since engine start > 2 minute(s)
- Idling > 15 second(s)
- Engine temperature > 40 °C (104 °F)
- Ambient temperature > 4 °C (39 °F)
- Ambient pressure > 730 hPa (= location below approx 2,400 ... 2,700 m/7,874 ... 8,858 ft) (A060)
- All oxygen sensor(s) ready for operation
- None of the following faults stored: Oxygen sensor(s), tank vent valve output stage, intake manifold pressure sensor, tank ventilation pressure sensor
- Vehicle speed > 15 km/h (9 mph)
- The carbon canister is adequately flushed via the tank vent valve (see ⓘ).

Possible fault causes

- ♦ Tank vent valve faulty (stuck open or closed)
- ♦ Tank vent valve not installed or installed incorrectly
- ♦ Line between active carbon filter and tank vent valve leaking/trapped/blocked
- ♦ Purge-air line between tank vent valve and turbocharger (Venturi pump) leaking/trapped/blocked
- ♦ Check valve between tank vent valve and intake manifold faulty or installed backward
- ♦ Check valve between tank vent valve and turbocharger (Venturi pump) faulty or installed backward
- ♦ Carrier-flow hose (Venturi pump) on turbocharger leaking or constricted
- ♦ Venturi pump on turbocharger faulty or constricted

Fault setting condition(s)

- The fault is set if the pressure difference between the ambient pressure and the pressure in the tank vent line is too low when the tank vent valve is activated (depending on the respective engine operating point).

ⓘ INFORMATION

This is achieved, for example, by driving:

- from vehicle standstill
- with the highest load possible (no sport mode, wide open throttle without kickdown, all loads switched on, uphill driving, ...)
- in the M gate
- always in the highest possible gear
- at engine speeds < 2,300 rpm
- for approx. 5 seconds with tank vent valve open (L170)
- with boost pressure > 300 hPa above ambient pressure (P020 + A060)

P0503

Diagnostic information - DME (DFI) control unit

Vehicle speed PSM

Diagnostic conditions

- Driving
- None of the following faults stored: U0129

Possible fault causes

- ♦ Fault in PSM area

Fault setting condition(s)

- Vehicle speed > 350 km/h or 218 mph (measured value)

P0506

Diagnostic information - DME (DFI) control unit

Idle speed control at stop, lower limit value

Diagnostic conditions

- Engine temperature > 60 °C (140 °F)
- Idling > 20 second(s)
- Vehicle standstill
- Engine without load (A/C system off, steering straight ahead)
- Ambient pressure > 730 hPa (= location below approx 2,400 ... 2,700 m/7,874 ... 8,858 ft.)
⇒ See actual value [A060-Ambient pressure from DME \(measured\)](#)
- None of the following functions active: Tank ventilation, Secondary air injection
- None of the following faults stored: Speed, throttle potentiometer, engine temperature, intake air temperature, tank vent valve, tank ventilation system

Possible fault causes

- ♦ Fuel quality insufficient
- ♦ Air cleaner blocked/frozen
- ♦ Air guide ahead of throttle valve blocked
- ♦ Ancillary units sluggish
- ♦ Exhaust system blocked

Fault setting condition(s)

Idle speed deviation > 20 second(s)

P0507

Diagnostic information - DME (DFI) control unit

Idle speed control at stop, upper limit value

Diagnostic conditions

- Engine temperature > 60 °C (140 °F)
- Idling > 20 second(s)
- Vehicle standstill
- Engine without load (A/C system off, steering straight ahead)
- Ambient pressure > 730 hPa (= location below approx 2,400 ... 2,700 m/7,874 ... 8,858 ft.)
⇒ See actual value [A060-Ambient pressure from DME \(measured\)](#)
- None of the following functions active: Tank ventilation, Secondary air injection
- None of the following faults stored: Speed, throttle potentiometer, engine temperature, intake air temperature, tank vent valve, tank ventilation system

Possible fault causes

- ♦ Intake manifold leaking
- ♦ Positive crankcase ventilation leaking
- ♦ Throttle valve adjusting unit faulty

Fault setting condition(s)

Idle speed deviation > 20 second(s)

P050A

Diagnostic information - DME (DFI) control unit

Idle speed control after cold start at stop, upper limit value

Diagnostic conditions

- Engine temperature -10 °C (14 °F)
- Cold start (catalytic converter heating measure)
- Idling > 20 second(s)
- Vehicle standstill
- Engine without load (A/C system off, steering straight ahead)
- Ambient pressure > 730 hPa (= location below approx 2,400 ... 2,700 m/7,874 ... 8,858 ft.)
⇒ See actual value [A060-Ambient pressure from DME \(measured\)](#)
- None of the following functions active: Tank ventilation, Secondary air injection
- None of the following faults stored: Speed, throttle potentiometer, engine temperature, intake air temperature, tank vent valve, tank ventilation system

Possible fault causes

- ♦ Intake manifold leaking
- ♦ Positive crankcase ventilation leaking
- ♦ Throttle valve adjusting unit faulty

Fault setting condition(s)

- Idle speed deviation > 20 second(s) during cold start

P050B

Diagnostic information - DME (DFI) control unit

Ignition timing angle after cold start

INFORMATION

This fault entry is used for information purposes only; the actual cause of the fault is in another area.

♦ Significant engine running problems force the engine control system to leave the permitted ignition timing range in order to keep the engine running.

Diagnostic conditions

- Cold start
- Then driving

P050C

Diagnostic information - DME (DFI) control unit

Engine coolant temperature implausible

INFORMATION

The fault **P2081-Thermostat stuck open** is always prioritized during fault finding if it was entered in addition to fault P0116.

♦ The value **DTC-DFCC** is shown in the extended fault memory for the respective fault code.

♦ **If only** fault **P0116** was entered, **check** which **DTC-DFCC** is entered in the extended fault memory.

► If **DTC-DFCC 00EE78** is entered, perform the **Thermostat system test before starting fault finding**

► see PIWIS Tester ⇒ DME ⇒ Maintenance/repairs ⇒ System tests

Diagnostic conditions

- Perform Thermostat system test

or

- Perform **First drive** with:
 - Engine start at (measured) engine temperature (**T020**) less than 30 °C (86 °F)
 - Drive for more than 20 minutes
- **Then drive the vehicle a second time** with:
 - Engine start again at (measured) engine temperature (**T020**) higher than 74 °C (165.2 °F)
 - Drive with engine speed greater than 1,300 rpm
 - High and low cooling power
 - Then with vehicle at a standstill, leave the engine running at idle speed for more than 45 seconds

Possible fault causes

 The value **DTC-DFCC** is shown in the extended fault memory for the respective fault code.

- For **DTC-DFCC 00EE77**:
 - ♦ Engine coolant temperature sensor faulty
 - ♦ Resistor installed (instead of coolant temperature sensor)
- For **DTC-DFCC 00EE78**:
 - ♦ Thermostat faulty
 - ♦ Seal in thermostat housing faulty
 - ♦ Thermostat housing faulty
 - ♦ Air in the cooling system
 - ♦ Combination valve for cooling system bleeding not closed (service bow is up)
 - ♦ Combination valve for cooling system bleeding leaking
 - ♦ Engine coolant temperature sensor faulty

Fault setting condition(s)

- For **DTC-DFCC** 00EE77:
 - Change in engine coolant temperature of less than 2 °C (35.6 °F) at diagnostic conditions
- For **DTC-DFCC** 00EE78:
 - The measured engine coolant temperature is lower than the model temperature at diagnostic conditions

P050D

Diagnostic information - DME (DFI) control unit

Idle speed control after cold start at stop, lower limit value

Diagnostic conditions

- Engine temperature -10 °C (14 °F)
- Cold start (catalytic converter heating measure)
- Idling > 20 second(s)
- Vehicle standstill
- Engine without load (A/C system off, steering straight ahead)
- Ambient pressure > 730 hPa (= location below approx 2,400 ... 2,700 m/7,874 ... 8,858 ft.)
⇒ See actual value [A060-Ambient pressure from DME \(measured\)](#)
- None of the following functions active: Tank ventilation, Secondary air injection
- None of the following faults stored: Speed, throttle potentiometer, engine temperature, intake air temperature, tank vent valve, tank ventilation system

Possible fault causes

- ♦ Fuel quality insufficient
- ♦ Air cleaner blocked/frozen
- ♦ Air guide ahead of throttle valve blocked
- ♦ Ancillary units sluggish
- ♦ Exhaust system blocked

Fault setting condition(s)

Idle speed deviation > 20 second(s) during cold start

P0513

Diagnostic information - DME (DFI) control unit

Immobilizer code incorrect from front-end electronics

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Immobilizer code incorrect from front-end electronics

Fault effects

- ♦ Engine cannot be started

P0521

Diagnostic information - DME (DFI) control unit

Oil pressure sensor implausible

Diagnostic conditions

- Idling > 1 minute(s), then:
 - Engine speed > 3,010 rpm, then:
 - Switch off ignition and remove ignition key > 1 minute(s)

Possible fault causes

- ♦ Engine oil pressure too low
(oil level, oil thinning, dirt in oil circuit, oil circuit faulty)
- ♦ Oil pressure sensor faulty (incorrect trimming compared with ambient pressure sensor)

Fault setting condition(s)

- Engine oil pressure (O060) > 5.2 bar relative at engine speed < 1,000 rpm for more than 5 seconds
or
- Engine oil pressure (O060) < 0.1 bar relative at engine speed > 3,008 rpm for more than 5 seconds
or
- Deviation between engine oil pressure (O060) and ambient pressure (A060) > 0.4 bar in control unit run-on (after ignition off)

P0522

Diagnostic information - DME (DFI) control unit

Oil pressure sensor

Diagnostic conditions

- Engine start detected

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Oil pressure sensor faulty

P0523

Diagnostic information - DME (DFI) control unit

Oil pressure sensor

Diagnostic conditions

- Engine start detected

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Oil pressure sensor faulty

P0524

Diagnostic information - DME (DFI) control unit

Engine oil pressure too low

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Oil level too low
- ♦ Oil dilution
- ♦ Dirt in oil circuit
- ♦ Oil circuit faulty

Fault setting condition(s)

- Actual oil pressure (O060) is less than setpoint oil pressure (O050) for too long, depending on the oil temperature

Diagnostic information - DME (DFI) control unit

Intake camshaft adjustment, bank 1 after cold start

- during catalytic converter heating measure

INFORMATION

If only the fault for camshaft adjustment after cold start is entered, particular attention should be paid to causes such as viscous engine oil or components that stick when cold.

INFORMATION

- ♦ The catalytic converter heating measures take place for a limited time under certain conditions (cold start).
- ♦ These include dual injection, timing retardation, additional camshaft adjustment, ...

Diagnostic conditions

- Cold start with catalytic converter heating measure
- then engine speed 1,200 ... 6,000 rpm
- None of the following faults stored: P0010, P0020, P2088, P2089, P2092, P2093

Possible fault causes

- ♦ Engine oil pressure too low
⇒ (e.g. oil thickening, wrong oil quality, oil pressure control or oil pump sluggish/stuck)
- ♦ Solenoid valve for hydraulic camshaft adjustment sluggish/stuck
- ♦ Camshaft adjustment sluggish/stuck
- ♦ Intake camshaft sluggish/stuck

Fault setting condition(s)

- Intake camshaft actual value/setpoint value deviation too high

Diagnostic information - DME (DFI) control unit

Intake camshaft adjustment, bank 2 after cold start

- during catalytic converter heating measure

INFORMATION

If only the fault for camshaft adjustment after cold start is entered, particular attention should be paid to causes such as viscous engine oil or components that stick when cold.

INFORMATION

- ♦ The catalytic converter heating measures take place for a limited time under certain conditions (cold start).
- ♦ These include dual injection, timing retardation, additional camshaft adjustment, ...

Diagnostic conditions

- Cold start with catalytic converter heating measure
- then engine speed 1,200 ... 6,000 rpm
- None of the following faults stored: P0010, P0020, P2088, P2089, P2092, P2093

Possible fault causes

- ♦ Engine oil pressure too low
⇒ (e.g. oil thickening, wrong oil quality, oil pressure control or oil pump sluggish/stuck)
- ♦ Solenoid valve for hydraulic camshaft adjustment sluggish/stuck
- ♦ Camshaft adjustment sluggish/stuck
- ♦ Intake camshaft sluggish/stuck

Fault setting condition(s)

- Intake camshaft actual value/setpoint value deviation too high

Diagnostic information - DME (DFI) control unit

Fuel high pressure after cold start

INFORMATION

- ♦ Faults in the fuel high-pressure injection system area are frequently caused by defects in the pre-delivery circuit (fuel low pressure).
- ♦ Therefore, the pre-delivery circuit must be checked first in case of problems in the area of the fuel high-pressure injection system.

INFORMATION

- ♦ The catalytic converter heating measures take place for a limited time under certain conditions after the engine is started.
- ♦ These include dual injection, ignition timing retardation, additional camshaft adjustment, ...

Diagnostic conditions

- Engine start (cold start + catalytic converter heating measure)
- Engine temperature > -10 °C (14 °F)
- Engine running > 5 second(s)

Possible fault causes

- ♦ Pre-delivery quantity or pressure in the fuel low-pressure circuit too low
- ♦ Quantity control valve faulty (in fuel high-pressure pump)
- ♦ Fuel high-pressure pump faulty
- ♦ Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see  below)

Fault setting condition(s)

Difference between fuel high pressure actual value and setpoint value too high during catalytic converter heating measure.

INFORMATION

Example

- ♦ Vehicle speed approx. 140 km/h (86 mph)
- ♦ Little or no acceleration although accelerator pedal is floored
- ♦ Actual values, or values in extended fault memory when fault was entered or when lack of power was detected, showed:
 - ♦ Fault entry P2414
 - ♦ Oxygen sensing, bank 1 = 25.0% (enrichment)
 - ♦ Fuel trim adaptation (load-dependent), bank 1 = 14.844%
 - ♦ L070_Oxygen sensor (LSU) voltage ahead of cat. con. 1 = 2.98828 V
 - ♦ L090_Oxygen sensor voltage behind cat. con. 1 = 0.00000 V

Explanation

The inlet for the fuel high-pressure rail behind the high-pressure pump is blocked either fully or partially by foreign bodies.

OR

The connection between the two fuel high-pressure rails is blocked either fully or partially by foreign bodies.

i The inlet of the respective rail is constricted by a restrictor, which is used to dampen pressure peaks.

- ♦ The fuel high-pressure sensor, which is fitted at the end of the lines, measures a pressure increase that is too slow, or a rapid drop in pressure when the fuel injectors open.
- ♦ As a result, fuel high pressure that is too high is set in the rail AHEAD OF the blockage, while at the same time, there is insufficient pressure and volume BEHIND the blockage.
- ♦ The oxygen sensors of the affected bank measure a large oxygen surplus, while at the same time, fuel trim adaptation tries to enrich the mixture as much as possible (extension of the injection time).

Consequences

- ♦ The following faults can be entered individually, in combination, or might not be entered at all:
 - ♦ Mixture control fault
 - ♦ Load detection fault
 - ♦ Misfiring
 - ♦ Oxygen sensor fault

AND

- ♦ Lack of power in upper load range can be reproduced, but is not necessarily always present.

OR

- ♦ The engine can stall again while starting.

Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.

Diagnostic information - DME (DFI) control unit

Bank 1 exhaust camshaft adjustment after cold start

- during catalytic converter heating measure

INFORMATION

If only the fault for camshaft adjustment after cold start is entered, particular attention should be paid to causes such as viscous engine oil or components that stick when cold.

INFORMATION

- ♦ The catalytic converter heating measures take place for a limited time under certain conditions after the engine is started.
- ♦ These include dual injection, timing retardation, additional camshaft adjustment, ...

Diagnostic conditions

- Cold start with catalytic converter heating measure
- then engine speed 1,200 ... 6,000 rpm
- None of the following faults stored: P0013, P0023, P2090, P2091, P2094, P2095

Possible fault causes

- ♦ Engine oil pressure too low
⇒ (e.g. oil thickening, wrong oil quality, oil pressure control or oil pump sluggish/stuck)
- ♦ Solenoid valve for hydraulic camshaft adjustment sluggish/stuck
- ♦ Camshaft adjustment sluggish/stuck
- ♦ Exhaust camshaft sluggish/stuck

Fault setting condition(s)

- Exhaust camshaft actual value/setpoint value deviation too high

Diagnostic information - DME (DFI) control unit

Bank 1 exhaust camshaft adjustment after cold start

- during catalytic converter heating measure

INFORMATION

If only the fault for camshaft adjustment after cold start is entered, particular attention should be paid to causes such as viscous engine oil or components that stick when cold.

INFORMATION

- ♦ The catalytic converter heating measures take place for a limited time under certain conditions after the engine is started.
- ♦ These include dual injection, timing retardation, additional camshaft adjustment, ...

Diagnostic conditions

- Cold start with catalytic converter heating measure
- then engine speed 1,200 ... 6,000 rpm
- None of the following faults stored: P0013, P0023, P2090, P2091, P2094, P2095

Possible fault causes

- ♦ Engine oil pressure too low
⇒ (e.g. oil thickening, wrong oil quality, oil pressure control or oil pump sluggish/stuck)
- ♦ Solenoid valve for hydraulic camshaft adjustment sluggish/stuck
- ♦ Camshaft adjustment sluggish/stuck
- ♦ Exhaust camshaft sluggish/stuck

Fault setting condition(s)

Exhaust camshaft actual value/setpoint value deviation too high

Diagnostic information - DME (DFI) control unit

Brake vacuum sensor implausible

Diagnostic conditions

- 25 times while driving under the following conditions:
 - Driving > 40 second(s) with:
 - Brake pedal not pressed
 - Then driving with brake pedal pressed, at:
 - Vehicle speed > 10 km/h (6 mph)
 - Brake pressure > 10 bar
 - Then driving > 40 second(s) with:
 - Brake pedal not pressed
 - Then driving with brake pedal pressed, at:
 - Vehicle speed > 10 km/h (6 mph)
 - Brake pressure > 10 bar

Possible fault causes

- ♦ Vacuum system faulty
- ♦ Brake booster faulty
- ♦ Vacuum sensor faulty

Fault setting condition(s)

- The vacuum measured by the sensor is maintained despite braking.

Fault effects

- ♦ Start Stop system not active